

Network Theorems:

Introduction:

Any complicated network i.e. several sources, multiple resistors are present if the single element response is desired then use the network theorems. Network theorems are also can be termed as network reduction techniques. Each and every theorem got its importance of solving network. Let us see some important theorems with DC and AC excitation with detailed procedures.

Thevenin's Theorem and Norton's theorem (Introduction) :

Thevenin's Theorem and Norton's theorem are two important theorems in solving Network problems having many active and passive elements. Using these theorems the networks can be reduced to simple equivalent circuits with one active source and one element. In circuit analysis many a times the current through a branch is required to be found when it's value is changed with all other element values remaining same. In such cases finding out every time the branch current using the conventional mesh and node analysis methods is quite awkward and time consuming. But with the simple equivalent circuits (with one active source and one element) obtained using these two theorems the calculations become very simple. Thevenin's and Norton's theorems are dual theorems.

Thevenin's Theorem Statement:

Any linear, bilateral two terminal network consisting of sources and resistors(Impedance), can be replaced by an equivalent circuit consisting of a voltage source in series with a resistance (Impedance). The equivalent voltage source V_{Th} is the open circuit voltage looking into the terminals(with concerned branch element removed) and the equivalent resistance R_{Th} while all sources are replaced by their internal resistors at ideal condition i.e. voltage source is short circuit and current source is open circuit.

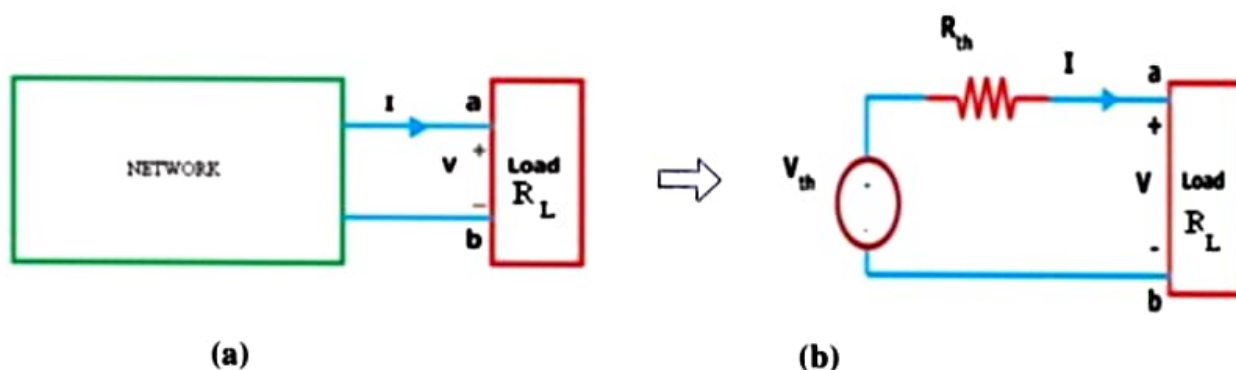


Figure (a) shows a simple block representation of a network with several active / passive elements with the load resistance R_L connected across the terminals 'a & b' and figure (b) shows the Thevenin's equivalent circuit with V_{Th} connected across R_{Th} & R_L .

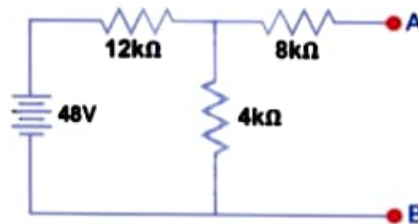
Main steps to find out V_{Th} and R_{Th} :

1. The terminals of the branch/element through which the current is to be found out are marked as say a & b after removing the concerned branch/element
2. Open circuit voltage V_{OC} across these two terminals is found out using the conventional network mesh/node analysis methods and this would be V_{Th} .
3. Thevenin's resistance R_{Th} is found out by the method depending upon whether the network contains dependent sources or not.
 - a. With dependent sources: $R_{Th} = V_{oc} / I_{sc}$
 - b. Without dependent sources : $R_{Th} =$ Equivalent resistance looking into the concerned terminals with all voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited)
4. Replace the network with V_{Th} in series with R_{Th} and the concerned branch resistance (or) load resistance across the load terminals (A&B) as shown in below fig.(a)

Example: Find V_{TH} , R_{TH} and the load current and load voltage flowing through R_L resistor as shown in fig. by using Thevenin's Theorem?

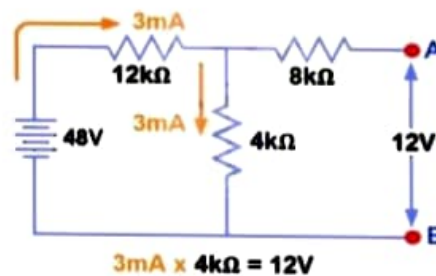
Solution:

The resistance R_L is removed and the terminals of the resistance R_L are marked as A & B as shown in the fig. (1)

**Fig.(1)**

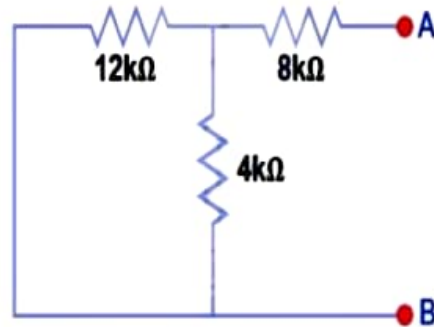
Calculate / measure the Open Circuit Voltage. This is the Thevenin Voltage (V_{TH}). We have already removed the load resistor from fig.(a), so the circuit became an open circuit as shown in fig (1). Now we have to calculate the Thevenin's Voltage. Since 3mA Current flows in both 12kΩ and 4kΩ resistors as this is a series circuit because current will not flow in the 8kΩ resistor as it is open. So 12V ($3mA \times 4k\Omega$) will appear across the 4kΩ resistor. We also know that current is not flowing through the 8kΩ resistor as it is open circuit, but the 8kΩ resistor is in parallel with 4k resistor. So the same voltage (i.e. 12V) will appear across the 8kΩ resistor as 4kΩ resistor. Therefore 12V will appear across the AB terminals.

So, $V_{TH} = 12V$

**Fig (2)**

All voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short

circuit and ideal current sources open circuited) as shown in fig.(3)



Fig(3)

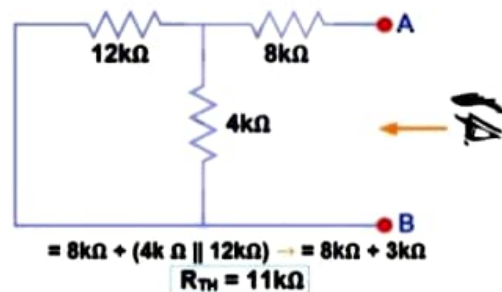
Calculate /measure the Open Circuit Resistance. This is the Thevenin's Resistance (R_{TH}) We have Reduced the 48V DC source to zero is equivalent to replace it with a short circuit as shown in figure (3) We can see that $8k\Omega$ resistor is in series with a parallel connection of $4k\Omega$ resistor and $12k\Omega$ resistor. i.e.:

$$8k\Omega + (4k\Omega \parallel 12k\Omega) \dots \dots (|| = \text{in parallel with})$$

$$R_{TH} = 8k\Omega + [(4k\Omega \times 12k\Omega) / (4k\Omega + 12k\Omega)]$$

$$R_{TH} = 8k\Omega + 3k\Omega$$

$$R_{TH} = 11k\Omega$$



Fig(4)

Connect the R_{TH} in series with Voltage Source V_{TH} and re-connect the load resistor across the load terminals(A&B) as shown in fig (5) i.e. Thevenin's circuit with load resistor. This is the Thevenin's equivalent circuit.

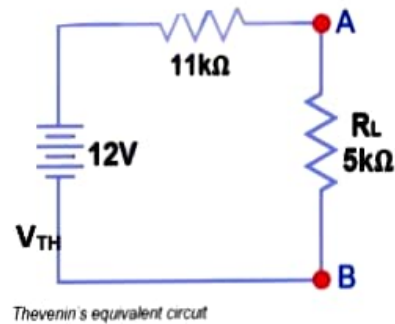


Fig (5)

Now apply Ohm's law and calculate the load current from fig 5.

$$I_L = V_{TH} / (R_{TH} + R_L) = 12V / (11k\Omega + 5k\Omega) = 12/16k\Omega$$

$$I_L = 0.75mA$$

$$\text{And } V_L = I_L \times R_L = 0.75mA \times 5k\Omega$$

$$V_L = 3.75V$$

Norton's Theorem Statement:

Any linear, bilateral two terminal network consisting of sources and resistors (Impedance), can be replaced by an equivalent circuit consisting of a current source in parallel with a resistance (Impedance), the current source being the short circuited current across the load terminals and the resistance being the internal resistance of the source network looking through the open circuited load terminals.

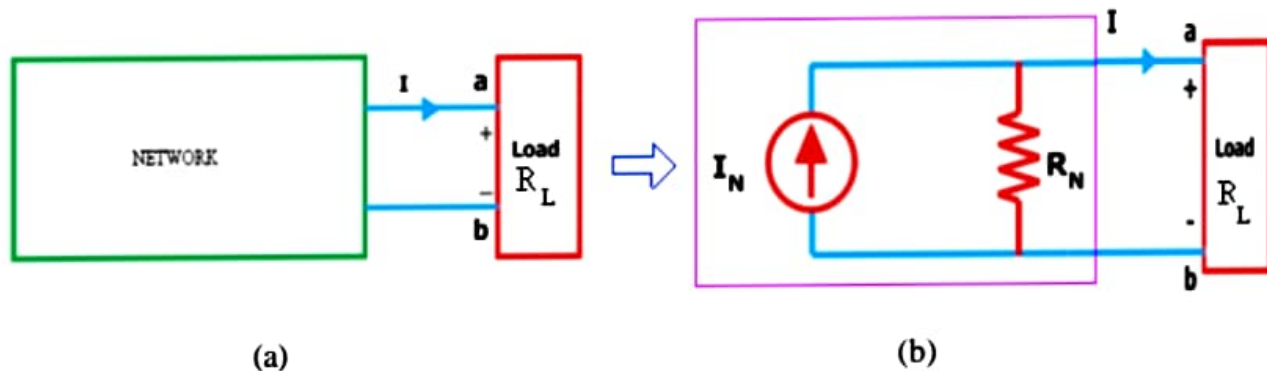
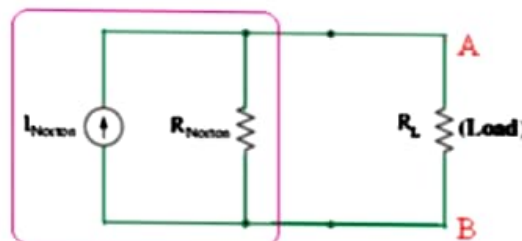


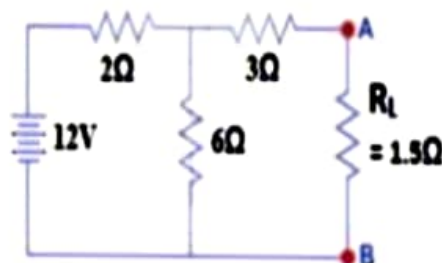
Figure (a) shows a simple block representation of a network with several active / passive elements with the load resistance R_L connected across the terminals 'a & b' and figure (b) shows the **Norton equivalent circuit** with I_N connected across R_N & R_L .

Main steps to find out I_N and R_N :

- The terminals of the branch/element through which the current is to be found out are marked as say **a & b** after removing the concerned branch/element.
- Open circuit voltage V_{oc} across these two terminals and I_{sc} through these two terminals are found out using the conventional network mesh/node analysis methods and they are same as what we obtained in Thevenin's equivalent circuit.
- Next **Norton resistance R_N** is found out depending upon whether the network contains dependent sources or not.
 - a) With dependent sources: $R_N = V_{oc} / I_{sc}$
 - b) Without dependent sources : $R_N =$ Equivalent resistance looking into the concerned terminals with all voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited)
- Replace the network with I_N in parallel with R_N and the concerned branch resistance across the load terminals(A&B) as shown in below fig



Example: Find the current through the resistance R_L (1.5Ω) of the circuit shown in the figure (a) below using Norton's equivalent circuit.



Fig(a)

Solution: To find out the Norton's equivalent ckt we have to find out $I_N = I_{sc}$, $R_N = V_{oc} / I_{sc}$. Short the 1.5Ω load resistor as shown in (Fig 2), and Calculate / measure the Short Circuit Current. This is the Norton Current (I_N).

We have shorted the AB terminals to determine the Norton current, I_N . The 6Ω and 3Ω are then in parallel and this parallel combination of 6Ω and 3Ω are then in series with 2Ω . So the Total Resistance of the circuit to the Source is:-

$$2\Omega + (6\Omega \parallel 3\Omega) \dots \dots (\parallel = \text{in parallel with})$$

$$R_T = 2\Omega + [(3\Omega \times 6\Omega) / (3\Omega + 6\Omega)]$$

$$R_T = 2\Omega + 2\Omega$$

$$R_T = 4\Omega$$

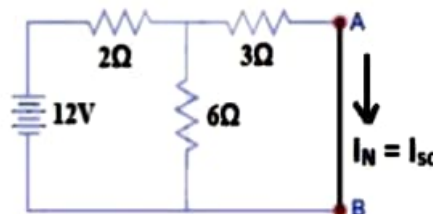
$$I_T = V / R_T$$

$$I_T = 12V / 4\Omega = 3A..$$

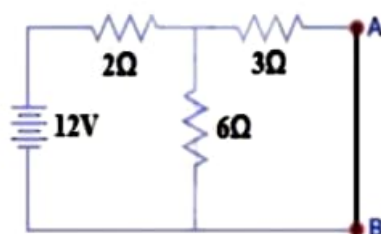
Now we have to find $I_{SC} = I_N$... Apply CDR... (Current Divider Rule)...

$$I_{SC} = I_N = 3A \times [(6\Omega / (3\Omega + 6\Omega))] = 2A.$$

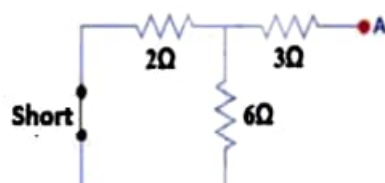
$$I_{SC} = I_N = 2A.$$



Fig(3)



All voltage & current sources replaced by their internal impedances (i.e. ideal voltage sources short circuited and ideal current sources open circuited) and Open Load Resistor. as shown in fig.(4)



Fig(4)

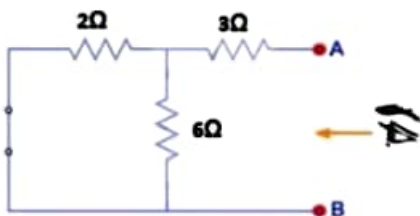
Calculate /measure the Open Circuit Resistance. This is the Norton Resistance (R_N) We have Reduced the 12V DC source to zero is equivalent to replace it with a short circuit as shown in fig(4), We can see that 3Ω resistor is in series with a parallel combination of 6Ω resistor and 2Ω resistor. i.e.:

$$3\Omega + (6\Omega \parallel 2\Omega) \dots \dots (\parallel = \text{in parallel with})$$

$$R_N = 3\Omega + [(6\Omega \times 2\Omega) / (6\Omega + 2\Omega)]$$

$$R_N = 3\Omega + 1.5\Omega$$

$$R_N = 4.5\Omega$$



Fig(5)

Connect the R_N in Parallel with Current Source I_N and re-connect the load resistor. This is shown in fig (6) i.e. Norton Equivalent circuit with load resistor.



Fig(6)

Now apply the Ohm's Law and calculate the load current through Load resistance across the terminals A&B. Load Current through Load Resistor is

$$I_L = I_N \times [R_N / (R_N + R_L)]$$

$$I_L = 2A \times (4.5\Omega / 4.5\Omega + 1.5k\Omega)$$

$$= 1.5A \quad I_L = 1.5A$$